PROJECT DEVELOPMENT PHASE

Business Questions • Tableau Visualisations • Feature Demonstration • Website Integration

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Course	Bachelor of Computer Applications (BCA)	**Semester**	Second Year / 3rd Semester
College	MEDI-CAPS University	**Project Type**	Individual Project

Global Fuel Prices and Taxation Analysis (2020–2026)

Purpose: To learn the concepts of Tableau and data analytics.

1. Business Questions & Visualisation Report

The project uses a historical fuel-pricing and taxation dataset to support descriptive comparisons across countries, regions, subsidy levels and income levels. The three dashboards are designed around policy analysis, supply-chain planning, and economic-development exploration.

Business question	Visualisation / purpose	Dashboard
Which countries have the highest average fuel-tax percentages?	Horizontal bar chart ranking countries by average tax percentage.	Fuel Taxation & Subsidy Policy
How does average petrol price vary across regions?	Regional bar chart with a region legend.	Fuel Taxation & Subsidy Policy
What relationship can be explored between taxation and petrol prices?	Scatter plot to compare association; it does not prove causation.	Fuel Taxation & Subsidy Policy
How do average fuel prices differ by subsidy level?	Bar chart grouped by subsidy level.	Fuel Taxation & Subsidy Policy
Which countries are in the top 10 for average petrol price, and how do petrol and diesel compare?	Ranked stacked bar comparison of petrol and diesel.	Global Fuel Prices & Supply Chain Planning
Which countries are in the bottom 10 for average petrol price, and how do petrol and diesel compare?	Ranked stacked bar comparison of petrol and diesel.	Global Fuel Prices & Supply Chain Planning
How have Brent crude and fuel prices changed over time in the dataset?	Time-series line chart for Brent crude, petrol and diesel prices.	Global Fuel Prices & Supply Chain Planning
How do average petrol prices and tax percentages compare across income levels?	Bar charts for petrol price and average tax percentage by income level.	Fuel Prices, Taxation & Economic Development

Scope note: The charts describe patterns in the available CSV data (January 2020–April 2026). They do not provide live market prices or establish cause-and-effect relationships.

2. Dashboard Development & Feature Demonstration

The supplied project reference documents three scenario dashboards and records the core dashboard, chart/legend, and publishing features as implemented and demonstrable. The table below summarises that evidence without claiming unverified performance testing.

Feature	Status	What it provides / limitation
Scenario 1 — Fuel Taxation & Subsidy Policy	Implemented; demonstrated	Country tax rankings, petrol prices by region, taxation-versus-petrol scatter plot, and fuel prices by subsidy level. Scatter plot indicates association only.
Scenario 2 — Global Fuel Prices & Supply Chain Planning	Implemented; demonstrated	Top/bottom country comparisons ranked by average petrol price, petrol/diesel comparison, regional petrol prices and historical Brent crude/fuel trends.
Scenario 3 — Fuel Prices, Taxation & Economic Development	Implemented; demonstrated	Income-level comparisons for petrol prices and tax percentages, with Income Level, Region and Subsidy Level controls visible in the supplied screenshot.
Visual comparison and legends	Implemented; demonstrated	Charts and legends support comparisons across countries, regions, fuel measures, tax percentages and subsidy categories. Interpret legends using the labels shown on each dashboard.
Published Tableau Public workbook	Published; demonstrated	The workbook can be opened in a browser to access the project dashboards. Use Disha's project link below rather than the original author's link in the source template.
Live refresh / database integration / separate Tableau Story	Not claimed as implemented	These are future extensions in the source reference. The project uses a CSV connected directly in Tableau; no live feed, separate SQL database, or formal load test is claimed.

Feature implementation summary

Tracked items	Implemented	Demonstrated	Recorded rate
6	5	5	83.3% (5 of 6 checklist items)

Counting note: 83.3% is a simple count in the adapted checklist, not a measured delivery or platform-performance score.

3. Website Development & Integration

A web interface complements the Tableau workbook by providing a single access point for the project overview and interactive dashboards. The source project reference identifies the following stack and deployment approach.

Technology	Role in the project
Python	Programming language used for the web application.
Flask	Python web framework used to serve the website.
HTML and CSS	Structure and visual styling of website content.
Tableau Public	Hosts the published interactive visualisations embedded in the website.
GitHub	Stores the website source code in the project repository.
Render	Deployment and online hosting platform for the web application.

Project access

Tableau Public Story	Open Disha's published Tableau Story
Live project website	global-fuel-prices-taxation-analysis.onrender.com
GitHub repository	DishaPatidar11/global-fuel-prices-taxation-analysis

Development notes & limitations

• The dataset was connected from a CSV file directly in Tableau Public; a separate SQL database is not claimed.
• The dashboards support descriptive comparisons. Correlation or visual association should not be interpreted as causation.
• No live data feed, formal user study, load test, mobile responsiveness test, or separate Tableau Story beyond the provided project link is claimed as independently verified.
• Rankings in the top/bottom 10 charts are based on average petrol price.

PROJECT PURPOSE	To learn the concepts of Tableau and data analytics by building scenario-based visualisations of global fuel prices, taxation and subsidy patterns.